Ioan Alexandru Popa

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Education

Aarhus University, **Master in Computer Science** — Aug 2026 - Jun 2028

- Relevant Coursework: Compilation (**OCaml**), Formal software verification (**Rocq**)

Politehnica University of Bucharest, **Bachelor in Computer Science** (Engineering) — Oct 2022 - Jul 2026

- Relevant Coursework: Programming (**C**, **Python**; **Assembly**; **Rust**), Data structures and algorithms, Object-oriented programming (**Java**), Operating systems, Databases (Oracle **SQL**), Cloud computing (**Kubernetes**)
- **Cumulative grades**: 9.22/10

Work experience

Nvidia — Helsinki, Finland
Software Infrastructure Engineering Intern — Jun 2025 - Sep 2025

- Reduced by **99**% the duration of action comparisons by implementing an **HTML** endpoint that compares 2 actions in an internal **Go** backend.
- Automated **4** steps of an OS image validation process using **Python** and **Bazel**.

Projects

Personal dotfiles management system (GitHub link) — Mar 2021 - May 2026

- Created a GitHub repository with configuration files for select programs, like `vim`, `zsh`, VSCode.
- Implemented **shell scripts** for **5** distro families that install a couple apps and configure a freshly installed **Linux** system with favourite configurations, and a **PowerShell** script that does this for **Windows**.

ThemeChanger (GitHub link) — Aug 2021 - May 2025

- Designed a Linux app in **Python** and **GTK3** that lets the user modify the mouse cursor, application icon, and widget themes and settings of **4** theme frameworks, even for unthemable libadwaita apps.
- Implemented live theme reloading using **6** desktop environment-specific mechanisms.
- Built a mechanism of showing instantly GTK3 theme and CSS changes in the app.

emscripten-functions (GitHub link) — Aug 2023 - Mar 2025

- Implemented raw **Rust** bindings for **emscripten** system functions. Emscripten is a compiler toolchain that allows C & C++ code to be ran on web pages using **WASM**.
- Built and documented Rust-friendly wrappers for **29** emscripten-specific functions.
- Over **12,000** downloads on `crates.io` for the `emscripten-functions` crate (the Rust-friendly wrappers), and over **14,000** for `emscripten-functions-sys` (the raw Rust bindings).

Volunteer activities

Undergraduate teaching assistant, Assembly programming — Feb 2024 - Jun 2024

- Held a weekly laboratory session for over **10** students that shows them the basics of **x86 Assembly**.
- Made a **Docker** image for lecture demos.

Open source contributions

- Added the ThemeChanger application to nixpkgs. (**Nix**) — Oct 2021 - Feb 2025
- Enhanced the tests suite of Lemmy and `rust-vmm/kvm-ioctls`. (**Rust**) — Jan 2024
- Fixed Romanian city names in OpenTTD. (**C++**) — Mar 2022